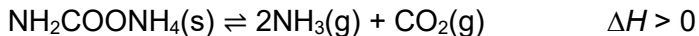


- 14 Ammonium carbamate, $\text{NH}_2\text{COONH}_4$, undergoes thermal decomposition.



A vessel containing only $\text{NH}_2\text{COONH}_4$ is heated to 250°C . The reaction reached equilibrium at time t_1 . Subsequently both the temperature and volume of the vessel are decreased, and the reaction established a new equilibrium at time t_2 .

Which statements are correct?

- 1 At t_2 , $P_{\text{NH}_3} : P_{\text{CO}_2}$ is 2 : 1.
- 2 The rate of the forward reaction at t_1 is the same as that at t_2 .
- 3 The degree of decomposition of $\text{NH}_2\text{COONH}_4$ at t_1 is smaller than that at t_2 .
- 4 Decreasing the volume of the vessel at constant temperature has no effect on the equilibrium partial pressures of NH_3 and CO_2 .

A 1 and 4

B 2 only

C 1 and 3

D 2 and 3